

Gold Money

When Do Gold Prices Matter for Exchange Rates?

Evidence from 21 years of daily data, across four numeraires

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Coverage: AUD, ZAR, CAD, CHF versus gold, measured in JPY, EUR, CHF and a currency basket

Sample: Daily, 4 January 2005 to 25 February 2026 (5,146 observations)

Asset classes: Gold (XAU), G10 and EM foreign exchange

Key Takeaways

- **The most robust finding: since 2020, daily gold returns and Australian dollar returns move together in every measuring currency tested.** The relationship is positive and statistically significant whether prices are measured in yen, euros, francs or a four-currency basket, and gold keeps its explanatory power after dollar and volatility controls in three of the four. No other currency in this study passes that test.
- **The dramatic headline numbers are the fragile way to tell the story.** The statistic that gold accounts for 63% of the variation in AUD comes from price levels measured in yen, and part of it reflects the yen's own weakness in 2022-24 rather than gold. In euro or basket terms the levels relationship fades or flips sign; the returns relationship survives everywhere. Levels flatter; returns carry the claim.
- **The South African rand fails in every specification.** With dollar and volatility controls, the rand's gold coefficient is negative in all four numeraire. Its historic gold-currency reputation does not survive any version of the test.
- **Formal break tests do not choose 2020.** A Quandt-Andrews search over all candidate dates places the dominant break for AUD in September 2008 (levels) and May 2011 (returns). The '2020s regime' framing in this report is a readable description of the recent era, not a statistically identified date.
- **Nothing ties any currency to gold in the long run.** Cointegration tests reject a stable equilibrium in every case. When gold and a currency drift apart, they can stay apart for years.
- **The relationship is not investable as a fixed rule.** An 11.5-year backtest earns 4.5% on AUD before costs, with a Sharpe ratio of 0.10 and a 9.9% maximum drawdown, and loses money on the other three currencies.

Summary assessment

Currency	Levels link 2020-26 (JPY)	Returns link 2020-26	Survives controls?	Classification
AUD	Strong (R2 0.63)	Positive in all 4 numeraire	Yes, in 3 of 4 numeraire	Gold-linked producer currency, regime-dependent
ZAR	Strong (R2 0.56)	Positive (common factor)	No, negative in all 4	Dollar-beta / risk currency
CAD	Strong (R2 0.60)	Positive	Yes (b 0.11, t 5.2)	Minor gold-linked producer
CHF	Strongest (R2 0.86)	Positive	Yes (JPY terms)	Safe-haven co-mover, zero mining exposure

The CHF row remains the cautionary tale: the strongest raw correlation in the sample belongs to a country with no gold mines. Correlation is not geology.

Positioning in the Literature

The building blocks of this study are well established. Chen and Rogoff (2003) established the commodity-currency framework for AUD, CAD and NZD. Sjaastad and Scacciavillani (1996) first connected gold to exchange rates through the denomination channel. Apergis (2014) examined gold and the Australian dollar directly. Erb and Harvey (2013) documented how often gold's investment roles are overstated. Relative to that work, this report contributes three things: it prices gold and currencies in several non-dollar numeraire to strip out the mechanical USD denomination effect and

quantify how much of the published relationship depends on that choice; it applies rolling-regime analysis and a formal structural-break search rather than assuming a stable relationship; and it closes the loop with a trading evaluation, so statistical significance is confronted with economic significance.

1. Research Design

You have probably heard the claim. Gold is up, so the Australian dollar should be up too, because Australia digs a lot of gold out of the ground. The same is said about the South African rand. It sounds sensible, and it gets repeated far more often than it gets tested. This report tests it.

Gold is a tricky asset to test, because it is not just a commodity. Copper is something you use; gold is mostly something you hold. People buy it when they are frightened, when interest rates are low, or when they distrust the dollar. Those same forces move every currency on earth. A naive correlation between gold and AUD therefore mixes up three different channels:

The income channel. When gold gets more expensive, a gold-exporting country earns more for the same shipments, so its currency should strengthen. This is the channel the market story assumes.

The denomination channel. Gold is priced in dollars. When the dollar weakens, the dollar price of gold rises and most currencies rise against the dollar at the same time, mechanically.

The common-factor channel. Fear and interest rates push gold and currencies around together without any real link between them. In a crisis, gold rises while the rand falls, and neither is causing the other.

1.1 Why these four currencies

Currency	Role	Rationale
AUD	Primary candidate	Australia is a top-three gold producer, gold is among its biggest exports, and the currency floats freely. If any currency should follow gold, it is this one.
ZAR	Historic candidate	South Africa was the world's largest producer for most of a century, but its mines have been shrinking for twenty years. It tests whether the label survives when the industry fades.
CAD	Control case	Canada mines plenty of gold but is a big, diversified exporter (oil, autos, metals). If CAD shows the same gold link as AUD, the link is probably global, not mining-related.
CHF	Look-alike detector	Switzerland mines no gold, but the franc and gold both attract safe-haven flows. A strong gold correlation here shows how large the look-alike effect can be.

Why not the other big producers? China consumes most of its own gold and manages its currency. Russia is under sanctions that drown out any commodity signal. Uzbekistan and the large West African producers run pegged or managed exchange rates, and a pegged currency cannot transmit a gold signal at all. Peru and Ghana are workable candidates and are flagged as natural extensions.

1.2 The numeraire problem, stated honestly

Because gold is priced in dollars, testing AUD/USD against gold-in-USD puts the dollar inside both series and manufactures correlation. The main analysis therefore measures everything in Japanese yen: Japan mines almost no gold, and the yen is deep and free-floating. But the yen is not perfectly neutral either. It has drivers of its own, including Bank of Japan policy, carry trades and safe-haven flows, and it was one of the weakest major currencies in 2022-24, which mechanically trended every yen-denominated price upward together.

Pricing in yen therefore substantially reduces the embedded USD mechanical correlation; it does not eliminate numeraire effects. For that reason, every key result in this report is re-estimated with the euro, the Swiss franc, and an equal-weight USD/EUR/JPY/GBP basket as alternative measuring sticks (Section 4), and conclusions are drawn only from what survives across all of them. An SDR-weighted basket is a natural further extension.

2. The Dollar Effect

Exhibit 1 is the single most important chart in this report. In dollar terms the rand looks strongly related to gold, though with a negative sign ($b = -0.57$, $R^2 = 0.53$). Re-priced in yen, the relationship mostly dissolves into two unrelated trends: a rising gold price and a rand that has been depreciating for twenty years.

The dollar effect: ZAR's 'gold link' depends on the numeraire

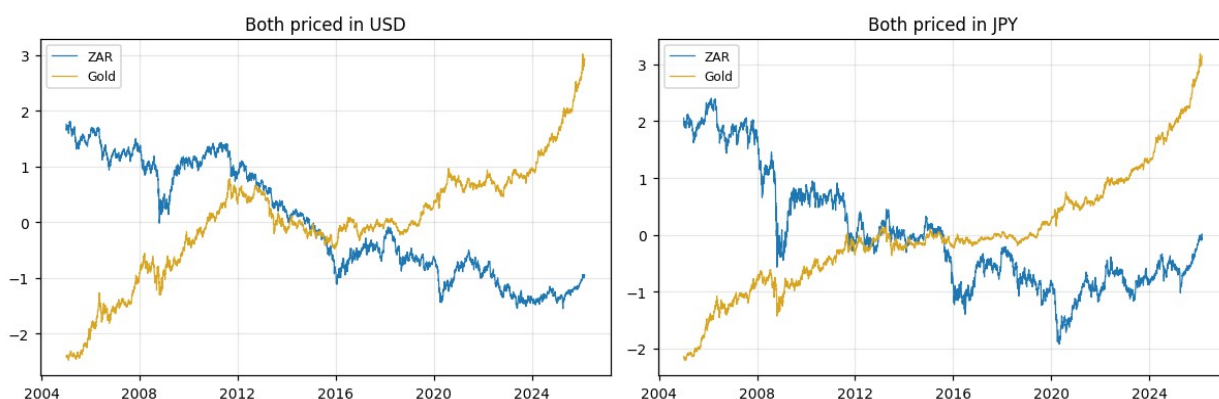


Exhibit 1: ZAR vs. gold, both priced in USD (left) and JPY (right). Normalized log levels.

Any analysis run purely in dollar terms inherits this distortion. It is probably the most common mistake in published gold-currency commentary.



Exhibit 2: Gold (USD/oz) vs. the broad USD index, 2006-2026. The index begins in January 2006, which is why the chart starts there.

3. Results by Era (Descriptive)

The sub-period table below uses hand-chosen splits and price levels, so it should be read as description rather than inference; the formal break analysis follows in Section 4. What it describes is how completely the apparent relationship changes across eras:

Period	AUD gold beta	Newey-West t	R2
2005-2012	-0.02	-1.3	0.00
2013-2019	-0.51	-3.2	0.11
2020-2026	+0.23	11.0	0.63

Gold accounts for 63% of the variation in AUD, measured in yen, during 2020-26. Two caveats belong right next to that number. First, these are regressions on trending, non-stationary price levels with no cointegration behind them (Section 5), so the R2 describes co-trending in that era rather than an equilibrium. Second, as Section 4 shows, the levels result is specific to the yen numeraire. The inference in this report rests on the robust t-statistics, the returns-based estimates, and the multi-numeraire checks, not on the levels R2.

With the broad dollar index and the VIX as controls (2006-2025, yen terms), the gold coefficient is +0.16 for AUD (t 12.2), +0.11 for CAD (t 5.2), +0.31 for CHF (t 24.1), and -0.15 for ZAR (t -3.3). The rand's apparent gold link is absorbed entirely by the dollar (dollar beta -1.22).

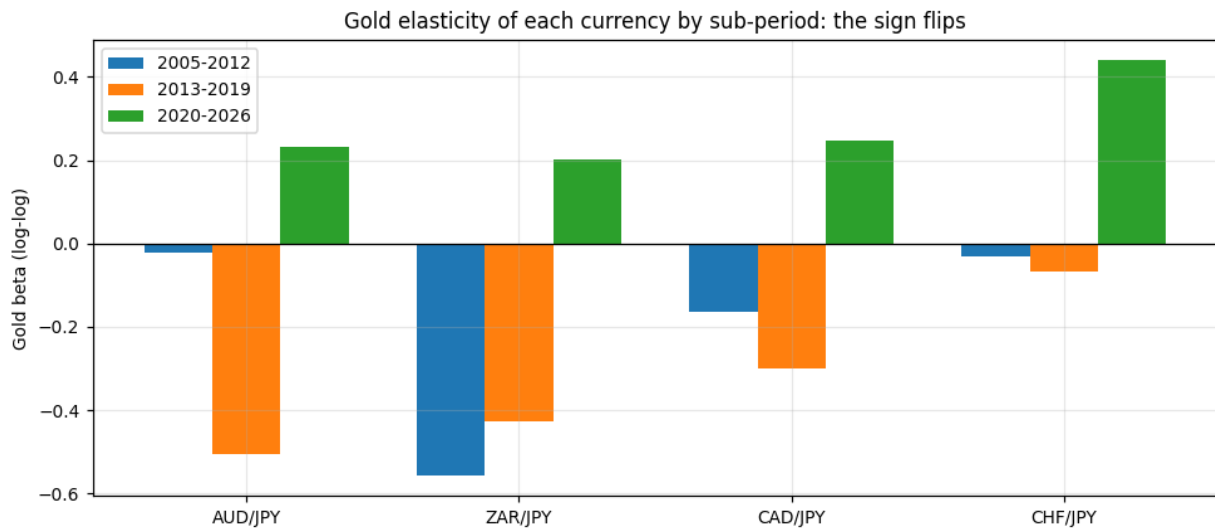


Exhibit 3: Gold sensitivity by sub-period. Negative or zero before 2020, positive everywhere after.

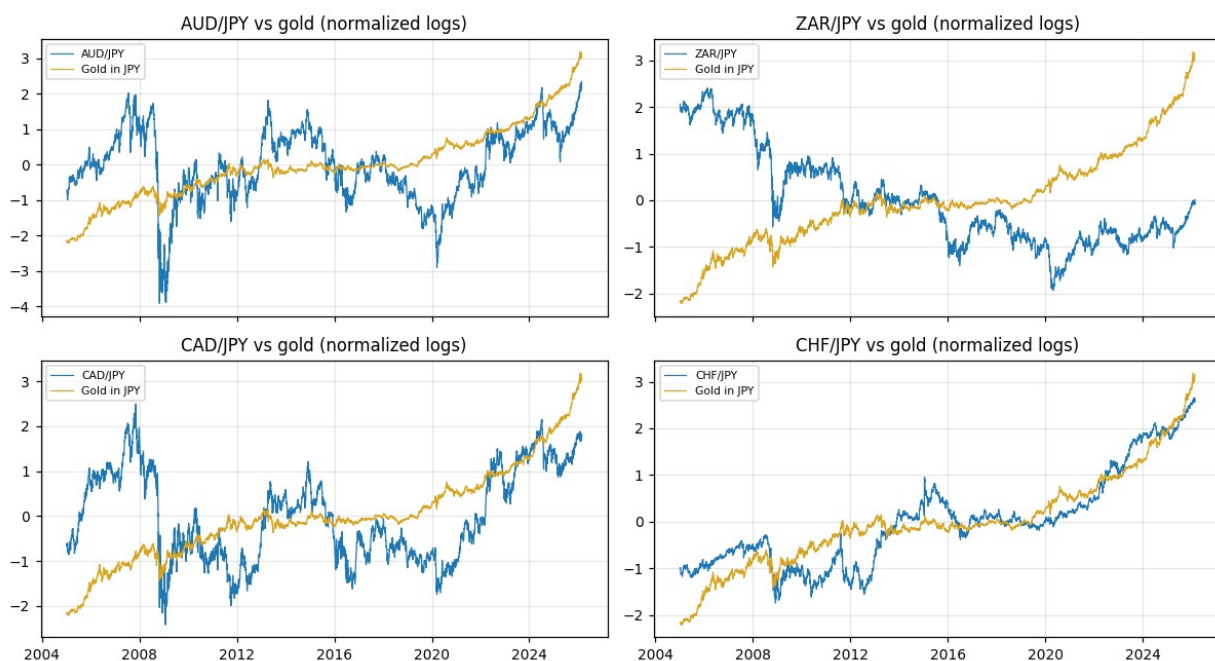


Exhibit 4: Each currency vs. gold in JPY terms, normalized log levels, 2005-2026.

4. Robustness: Numeraires and Formal Break Tests

4.1 Does the answer depend on the measuring stick?

The key AUD results, re-estimated in four measuring currencies:

AUD vs gold	JPY	EUR	CHF	Basket
Levels beta, 2020-26	+0.23 (t 11.0)	-0.11 (t -7.2)	-0.32 (t -11.4)	-0.08 (t -2.7)
Returns beta, 2020-26	+0.25 (t 7.9)	+0.10 (t 6.0)	+0.12 (t 5.2)	+0.43 (t 13.8)
Controls beta (full sample)	+0.16 (t 12.2)	+0.14 (t 7.7)	-0.17 (t -9.8)	+0.13 (t 8.9)

Read the first two rows together and the lesson is clear. The levels relationship is numeraire-dependent: strongly positive only in yen, partly because the weak yen of 2022-24 trended AUD/JPY and gold/JPY upward together. The returns relationship, by contrast, is positive and statistically significant in every numeraire, and the controls result holds in three of four. (The CHF-numeraire exception is expected: measuring against the franc subtracts the safe-haven factor that gold itself shares.) The defensible claim is therefore about returns: day to day, gold and the Australian dollar genuinely move together in this era, whatever you measure them in.

The rand needs no nuance. With controls, its gold coefficient is negative in every numeraire: -0.15 in yen, -0.22 in euros, -0.59 in francs, -0.24 in the basket. The rand is not a gold currency in any specification tested.

4.2 Is 2020 a real break date?

A Quandt-Andrews sup-F search over all candidate break dates (with 15% trimming) does not select 2020. For AUD the dominant single break falls in September 2008 in levels and May 2011 in returns; ZAR breaks in late 2015 (levels), CHF in April 2013. Two honest conclusions follow. First, the gold-FX relationship is unstable everywhere; the test statistics are enormous. Second, the 2020 split used in Section 3 is a readable description of the recent era, not a statistically identified regime date. A multi-break Bai-Perron analysis is the natural next step.

5. The Long Run and the Direction of Information

Is there a rubber band pulling gold and each currency back together? No. This is what cointegration tests measure, and they reject it everywhere: Engle-Granger fails for all four currencies (AUD comes closest at $p = 0.07$; ZAR is nowhere at $p = 0.88$), and the Johansen test agrees in every case. Where an adjustment speed can be estimated at all, closing half of a gap would take 190 to 350 trading days. When gold and a currency drift apart, nothing dependable pulls them back.

Who carries information about whom? Granger causality tests ask whether yesterday's gold returns contain statistically useful information for forecasting today's currency returns, and the reverse. That is predictive content, not economic causation, and the distinction matters. The p-values shown are the smallest across the five lags tested, which slightly flatters borderline results:

Direction	AUD	ZAR	CAD	CHF
Gold informs the currency (p-value)	0.001	0.39	0.011	0.39
The currency informs gold (p-value)	0.06	0.004	0.27	0.007

Gold carries forecasting information for the producer currencies (AUD, CAD): news lands in the gold market first and seeps into the currencies of the countries that mine it. For the rand and the franc, information flows the other way, because these currencies are barometers of global mood, and the mood moves gold. The direction of predictive content itself classifies the currency type.

Can gold explain currency moves on fresh data? Modestly. Using the same day's gold move, a gold-based model reduced forecast error, measured as mean squared error, by 9 to 12% relative to a random-walk benchmark, for all four currencies. The fine print: that test uses the same day's gold price, so it explains rather than predicts.

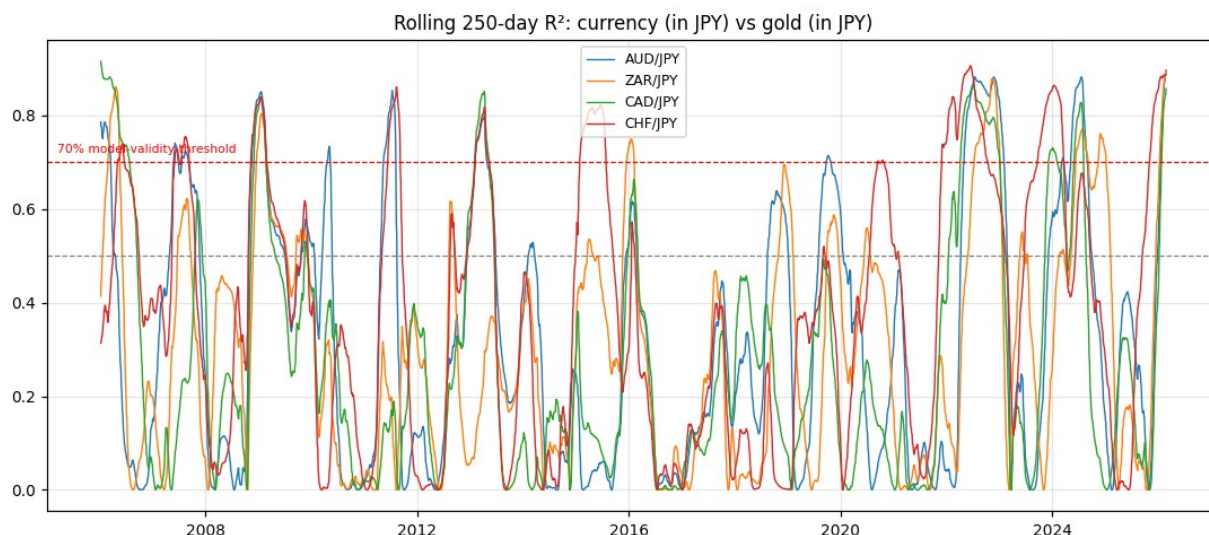


Exhibit 5: Rolling one-year fit (R^2) with gold. The link comes in episodes, above the 70% line only 7 to 16% of the time, clustered in 2008-09, 2011-13 and 2022-26. The rolling window needs a year of data, hence the 2006 start.

Exhibit 5 may be the most useful chart in the report. Being a gold currency is something a currency does occasionally, not something it is.

6. Strategy Backtest: Can You Make Money From This?

The test is deliberately simple. Fit a rolling 50-day model of the currency against gold. Only act when the model currently fits well (rolling R^2 of at least 50%). When the currency breaks two standard deviations out of its band, follow the move. Exit after ten days or at a 2% stop. The backtest runs from August 2014 (after a 50-day warm-up on data starting June 2014) through February 2026, before costs:

Currency	Trades	Total return	Win rate
AUD	43	+4.5%	53%
ZAR	45	-29.0%	36%
CAD	40	-5.0%	53%
CHF	45	-8.9%	44%

Institutional risk metrics for the AUD strategy, computed from its daily P&L series: annualized return 0.4%, annualized volatility 3.9%, Sharpe ratio 0.10, and a maximum drawdown of 9.9% over 11.5 years. Stated plainly: not investable as a fixed rule. The tuned +9.2% grid peak is in-sample optimization; returns are gross of costs, and at roughly ten basis points of average profit per trade, spreads and slippage in AUD/JPY would consume much of the edge; and the trade-level figures are sums of log returns on unit positions, not compounded portfolio returns. A real implementation would

require a regime filter that trades only inside the high-fit clusters of Exhibit 5, plus walk-forward validation, which is future work.

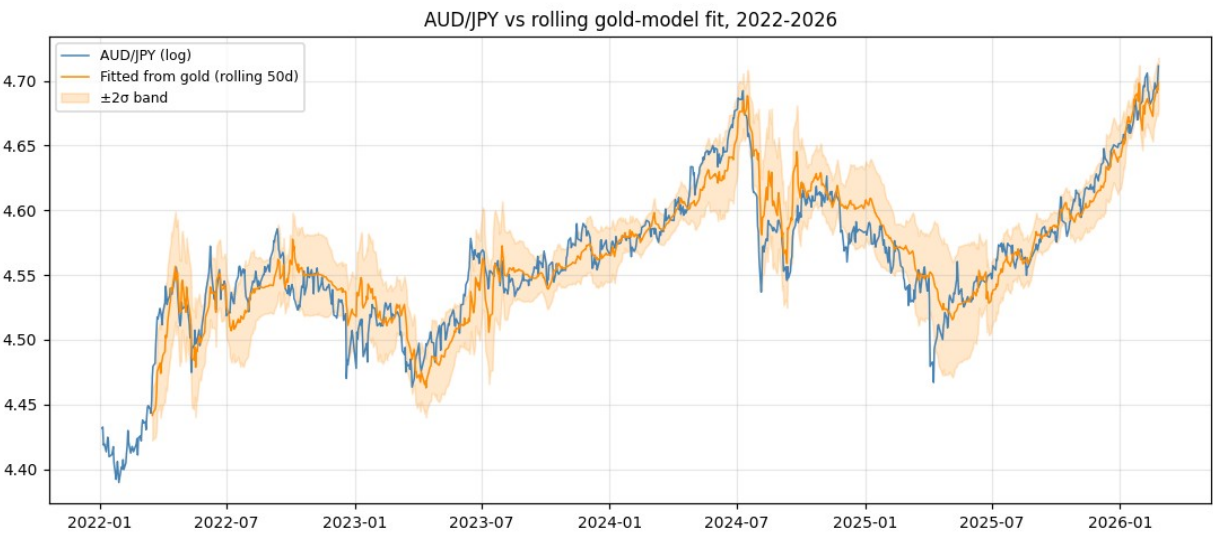


Exhibit 6: AUD/JPY vs. the rolling gold-model fit with 2-sigma bands, 2022-2026. Recent window shown for readability; the model runs on the full history.

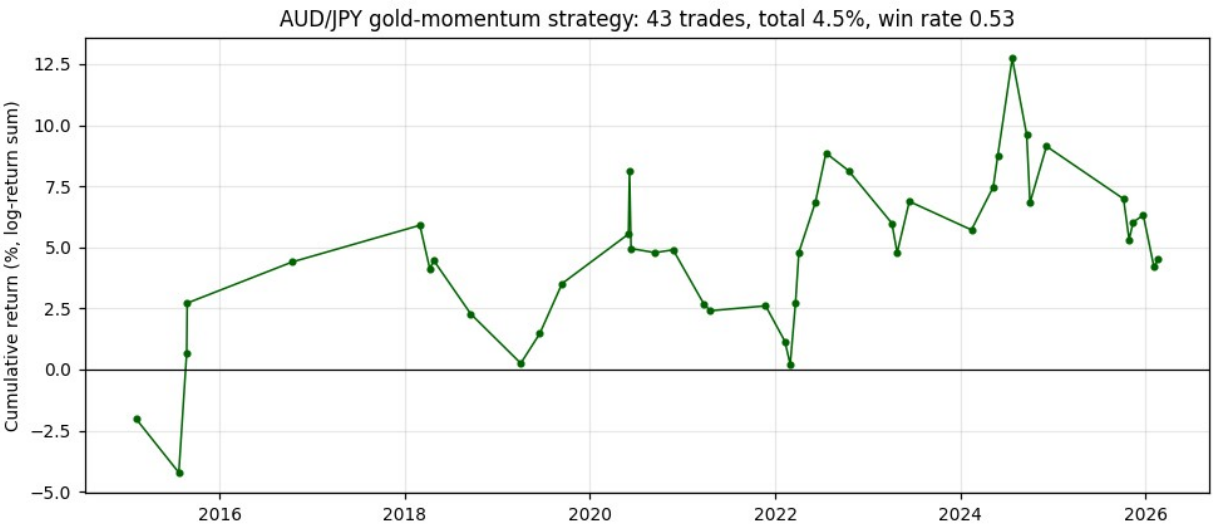


Exhibit 7: AUD strategy equity curve. 43 trades, +4.5% cumulative. X-axis points are trade exit dates.

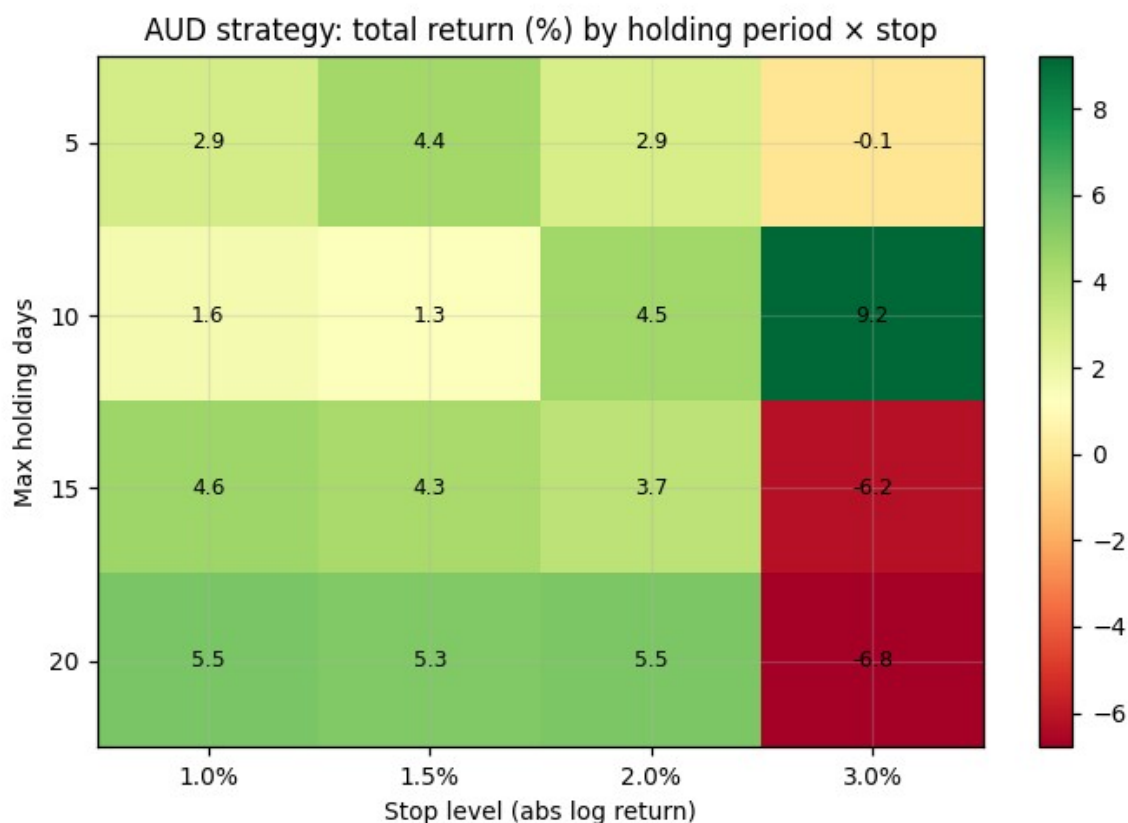


Exhibit 8: AUD total return across the holding-period and stop-level grid. Adjacent cells flip sign, the signature of a fragile edge.

7. What This Means

For investors. Do not read a gold chart as a currency forecast. The robust link is in daily returns, it is modest, and it switches on mainly in weak-dollar, high-fear eras. Watch the dollar and US real interest rates first, because they sit behind both sides of the chart.

For policymakers in gold-exporting countries. Gold booms can strengthen the currency when the global regime cooperates, as Australia after 2020 shows. But there is no long-run anchor, and South Africa shows how completely gold-currency status can fade as mines decline. A gold windfall deserves the same caution as any commodity windfall: save some, hedge some, and do not build the budget on it.

For researchers. The direction-of-information test cleanly separated producer currencies from risk currencies, an inexpensive and repeatable classification rule. Natural next steps: multi-break Bai-Perron dating, an SDR numeraire, interest-rate-differential and terms-of-trade controls, the Peruvian sol and Ghanaian cedi, and walk-forward validation of the trading rule.

8. Conclusion

When do gold prices matter for exchange rates? On the evidence here: in daily returns, for producer currencies, mostly in weak-dollar and high-anxiety eras, and never as a long-run anchor. The Australian dollar carries a genuine, robust gold link in this decade, one that survives every measuring stick and control this report could throw at it, provided the claim is stated in returns rather than in flattering price levels. The South African rand, the currency that made the phrase famous, fails the same tests in every form. The Swiss franc imitates a gold currency through shared safe-haven

demand, Swiss National Bank policy and European financial conditions, without a single mine, which is the clearest warning against reading correlation as geology.

The gold price is a mirror of the world's fears and its interest rates. The currencies of gold-mining countries are reflections in that mirror: real, but visible only when the light is right, and never fixed in place. Every confident claim that gold strength must lift a mining currency deserves the questions this report kept asking: measured in what, in levels or in returns, controlling for what, and in which era? The answers, more often than not, change the story.

Risk Factors and Limitations

- **Omitted variables.** Controls are limited to the broad dollar and the VIX. Interest-rate differentials, terms of trade, oil prices (especially for CAD), inflation surprises and current-account dynamics are omitted; part of the estimated gold effect may proxy for these.
- **Numeraires.** Four numeraires are tested; an SDR-weighted basket is left to future work. No numeraire is perfectly neutral.
- **Break analysis.** The Quandt-Andrews search identifies a single dominant break; multi-break Bai-Perron dating is future work.
- **Real rates.** No daily real-rate (TIPS) series was available in the free sources used.
- **Data coverage.** The broad dollar index covers 2006 to December 2025; the daily gold fix in the mirror used ends February 2026.
- **Backtest.** Results are before costs, without compounding, and without walk-forward validation.
- **Reproducibility.** The data mirrors update daily; to reproduce the exact numbers here, use the frozen files in the repository's data/raw folder.
- **Regime dependence.** Recent results are dominated by the 2020-26 era, which may not persist.

Appendix A: Data

Series	Source (mirror)	Range used
Gold PM fix, USD (daily)	LBMA via forex-centuries	2005 to 2026-02
Fed H.10 daily FX (local-per-USD)	datasets/exchange-rates	2005 to 2026-02
Broad USD index (DTWEXBGS)	FRED via forex-centuries	2006 to 2025-12
VIX daily close	CBOE via datasets/finance-vix	2005 to 2026-02
Gold monthly average	LBMA via datasets/gold-prices	context only

Crosses are built as numeraire units per unit of currency (for example $AUDJPY = (JPY/USD)/(AUD/USD)$); gold in a numeraire is the dollar fix multiplied by that numeraire's rate against the dollar. The basket numeraire is an equal-weight log combination of USD, EUR, JPY and GBP. All models use natural logs. The full daily panel, all model outputs and the backtest results are available in the project repository: github.com/maitrayee196/Gold-Prices-vs-Exchange-Rates.

Appendix B: Selected References

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About the Author

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